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Mata kuliah : Optimisasi

Tugas : Memecahkan Masalah

Soal

Consider the optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & 2x_1 + x_2 \geq 1 \\ & x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

Make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value.

(a) $f_0(x_1, x_2) = x_1 + x_2$

(b) $f_0(x_1, x_2) = -x_1 - x_2$

(c) $f_0(x_1, x_2) = x_1$

(d) $f_0(x_1, x_2) = \max\{x_1, x_2\}$

(e) $f_0(x_1, x_2) = x_1^2 + 9x_2^2$

Penyelesaian :

Sintak yang digunakan:

```
julia> using JuMP
```

```
julia> import Ipopt
```

```
julia> model = Model(Ipopt.Optimizer)
```

```
julia> @variable(model, x1 >=0)
```

```
julia> @variable(model, x2 >=0)
```

```
julia> @constraint(model, 2*x1 + x2 >=1)
```

```
julia> @constraint(model, x1 + 3*x2 >=1)
```

```
julia> @objective(model, Min, x1+x2)
```

```
julia> optimize!(model)
```

```
julia> @objective(model, Min, -x1-x2)
```

```
julia> optimize!(model)
julia> @objective(model, Min, x1)
julia> optimize!(model)
julia> @objective(model, Min, max(x1,x2))
julia> optimize!(model)
julia> @objective(model, Min, x1^2+9x2^2)
julia> optimize!(model)
```

Lampiran

(a) $f_0(x_1, x_2) = x_1 + x_2$

```

julia 1.10.5
Documentation: https://docs.julialang.org
Type "?" for help, "]" for pkg help.
Version 1.10.5 (2024-08-27)
Official https://julialang.org/ release

julia> using JuMP
julia> import Ipopt

julia> model = Model{Ipopt.Optimizer}
A JuMP Model
  solver: Ipopt
  objective_sense: FEASIBILITY_SENSE
  num_variables: 0
  num_constraints: 0
  Names registered in the model: none

julia> @variable(model, x1 >= 0)
x1
julia> @variable(model, x2 >= 0)
x2
julia> @constraint(model, 2*x1 + x2 >= 1)
2 x1 + x2 >= 1
julia> @constraint(model, x1 + 3*x2 >= 1)
x1 + 3 x2 >= 1
julia> @objective(model, Min, x1+x2)
x1 + x2
julia> optimize!(model)

*****
This program contains Ipopt, a library for large-scale nonlinear optimization.
Ipopt is released as open source code under the Eclipse Public license (EPL).
For more information visit https://github.com/coin-or/ipopt
*****

This is Ipopt version 3.14.16, running with linear solver MUMPS 5.7.3.

Number of monomers in equality constraint Jacobian... 0
Number of monomers in inequality constraint Jacobian... 4
Number of monomers in Lagrangian Hessian... 0

Total number of variables..... 2
  variables with only lower bounds: 2
  variables with lower and upper bounds: 0
Total number of equality constraints..... 0
  equality constraints with only lower bounds: 0
  equality constraints with only upper bounds: 0
  inequality constraints with lower and upper bounds: 2
  inequality constraints with only upper bounds: 0

Iteration   objective    inf_pr    inf_du lg(mu)  |d|  lg(r)  alpha_du  alpha_pr  ls
 0  1.9999998e-02  9.78e-01  9.76e-01 -1.0  0.00e+00 -  0.00e+00  0.00e+00  0
 1  6.1869308e-01  9.62e-01  1.77e+00 -1.7  3.20e-01 -  3.20e-01  7.90e-02  1
 2  6.0979157e-01  9.00e+00  1.40e+00 -1.7  1.40e-01 -  1.40e-01  1.00e+00  1
 3  6.1816884e-01  9.00e+00  2.00e-07 -1.7  1.43e-02 -  1.00e+00  1.00e+00  1
 4  6.0121372e-01  9.00e+00  1.00e-02 -3.8  4.21e-02 -  1.00e+00  5.01e-01  1
 5  6.0838377e-01  9.00e+00  1.50e-09 -3.8  2.35e-03 -  1.00e+00  1.00e+00  1
 6  6.0000000e-01  9.00e+00  1.00e-11 -5.7  7.52e-04 -  1.00e+00  1.00e+00  1
 7  6.0000000e-01  9.00e+00  2.51e-14 -8.6  8.41e-06 -  1.00e+00  1.00e+00  1

Number of iterations..... 7

(scaled)                (unscaled)
Objective..... 5.999999898211715e-01  5.999999898211715e-01
Dual infeasibility..... 2.590366264670015e-14  2.590366264670015e-14
Constraint violation..... 0.000000000000000e+00  0.000000000000000e+00
Variable bound violation..... 0.000000000000000e+00  0.000000000000000e+00
Complementarity..... 2.530515393211861e-09  2.530515393211861e-09
Overall NLP error..... 2.530515393211861e-09  2.530515393211861e-09

Number of objective function evaluations = 8
Number of objective gradient evaluations = 8
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 8
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 1
Number of Lagrangian Hessian evaluations = 1
Total seconds in IPOPT = 0.358

EXIT: Optimal Solution Found.

```

(b) $f_0(x_1, x_2) = -x_1 - x_2$

```

Julia 1.10.5
julia> @objective(model, Min, -x1-x2)
x1 - x2

julia> optimize!(model)
This is Ipopt version 3.14.16, running with linear solver MUMPS 5.7.3.

Number of nonzeros in equality constraint Jacobian...: 0
Number of nonzeros in inequality constraint Jacobian.: 4
Number of nonzeros in Lagrangian Hessian.....: 0

Total number of variables.....: 2
    variables with only lower bounds: 2
    variables with lower and upper bounds: 0
    variables with only upper bounds: 0
Total number of equality constraints.....: 0
Total number of inequality constraints.....: 2
    inequality constraints with only lower bounds: 2
    inequality constraints with lower and upper bounds: 0
    inequality constraints with only upper bounds: 0

iter   objective    inf_pr  inf_du lg(mu)  ||d||  lg(rg)  alpha_du  alpha_pr  ls
  0  -1.9999980e-02  9.70e-01  1.40e+00  -1.0  0.00e+00  -  0.00e+00  0.00e+00  0
  1  -2.6195585e-01  5.82e-01  1.03e+01  -1.0  3.60e-01  -  3.67e-02  4.06e-01f  1
  2  -6.6185945e-01  0.00e+00  7.27e-01  -1.0  2.54e-01  -  6.59e-01  1.00e+00f  1
  3  -1.2619819e+00  0.00e+00  7.15e-01  -1.7  1.28e+00  -  3.24e-01  1.00e+00f  1
  4  -1.0476432e+01  0.00e+00  1.01e+00  -1.7  1.79e+01  -  7.38e-02  1.00e+00f  1
  5  -3.6983762e+02  0.00e+00  1.02e+00  -1.7  7.15e+02  -  2.65e-02  1.00e+00f  1
  6  -7.4860748e+05  0.00e+00  5.19e-01  -1.7  1.48e+06  -  4.89e-04  1.00e+00f  1
  7  -1.6528634e+10  0.00e+00  4.68e-01  -1.7  1.57e+11  -  9.33e-06  2.10e-01f  1
  8  -1.7543300e+17  0.00e+00  8.03e-01  -1.7  3.51e+17  -  9.33e-08  1.00e+00f  1
  9  -4.7711077e+25  0.00e+00  8.03e-01  -1.7  3.72e+26  -  9.33e-10  2.55e-01f  1

Number of Iterations.....: 9

          (scaled)                (unscaled)
Objective.....: -4.771107697338632e+25  -4.771107697338632e+25
Dual infeasibility.....:  0.0291631090727145e-01  0.0291631090727145e-01
Constraint violation.....:  0.000000000000000e+00  0.000000000000000e+00
Variable bound violation:  0.000000000000000e+00  0.000000000000000e+00
Complementarity.....:  2.000000000383162e+08  2.000000000383162e+08
Overall NLP error.....:  2.000000000383162e+08  2.000000000383162e+08

Number of objective function evaluations = 10
Number of objective gradient evaluations = 10
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 10
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 1
Number of Lagrangian Hessian evaluations = 1
Total seconds in IPOPT = 0.290

```

(c) $f_0(x_1, x_2) = x_1$

```

Julia 1.10.5
julia> @objective(model, Min, x1)
x1

julia> optimize!(model)
This is Ipopt version 3.14.16, running with linear solver MUMPS 5.7.3.

Number of nonzeros in equality constraint Jacobian...: 0
Number of nonzeros in inequality constraint Jacobian.: 4
Number of nonzeros in Lagrangian Hessian.....: 0

Total number of variables.....: 2
    variables with only lower bounds: 2
    variables with lower and upper bounds: 0
    variables with only upper bounds: 0
Total number of equality constraints.....: 0
Total number of inequality constraints.....: 2
    inequality constraints with only lower bounds: 2
    inequality constraints with lower and upper bounds: 0
    inequality constraints with only upper bounds: 0

iter   objective    inf_pr  inf_du lg(mu)  ||d||  lg(rg)  alpha_du  alpha_pr  ls
  0  9.9999980e-03  9.70e-01  1.29e+00  -1.0  0.00e+00  -  0.00e+00  0.00e+00  0
  1  1.2895610e-01  6.74e-01  7.43e+00  -1.0  3.55e-01  -  3.73e-02  3.12e-01f  1
  2  4.1116772e-01  0.00e+00  1.90e-01  -1.0  2.90e-01  -  8.44e-01  1.00e+00f  1
  3  2.1022442e-01  0.00e+00  4.22e-01  -1.7  1.01e+00  -  4.12e-01  1.00e+00f  1
  4  2.1022343e-03  0.00e+00  1.49e-01  -1.7  1.66e+00  -  5.17e-01  6.83e-01f  1
  5  3.0598488e-03  0.00e+00  8.97e-03  -2.5  4.78e-02  -  9.68e-01  1.00e+00f  1
  6  1.5271189e-04  0.00e+00  3.24e-04  -3.8  4.18e-01  -  8.52e-01  1.00e+00f  1
  7  1.7061766e-06  0.00e+00  3.98e-06  -5.7  1.32e-02  -  9.95e-01  1.00e+00f  1
  8  -7.4914961e-09  0.00e+00  5.14e-09  -8.6  1.26e-02  -  9.97e-01  1.00e+00f  1

Number of Iterations.....: 8

          (scaled)                (unscaled)
Objective.....: -7.4914961310008418e-09  -7.4914961310008418e-09
Dual infeasibility.....:  5.1426611060079961e-09  5.1426611060079961e-09
Constraint violation.....:  0.000000000000000e+00  0.000000000000000e+00
Variable bound violation:  7.4914961310008418e-09  7.4914961310008418e-09
Complementarity.....:  2.5348402978582921e-09  2.5348402978582921e-09
Overall NLP error.....:  5.1426611060079961e-09  5.1426611060079961e-09

Number of objective function evaluations = 9
Number of objective gradient evaluations = 9
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 9
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 1
Number of Lagrangian Hessian evaluations = 1
Total seconds in IPOPT = 0.281

```

(d) $f_0(x_1, x_2) = \max\{x_1, x_2\}$

Julia 1.10.5

```
Julia> objective(model, Min, max(x1,x2))
max(x1, x2)

Julia> optimize!(model)

This is Ipopt version 3.14.16, running with linear solver MUMPS 5.7.3.
```

```
Number of nonzeros in equality constraint Jacobian...: 0
Number of nonzeros in inequality constraint Jacobian...: 4
Number of nonzeros in Lagrangian Hessian...: 3

Total number of variables...: 2
  variables with only lower bounds: 2
  variables with lower and upper bounds: 0
  variables with only upper bounds: 0
Total number of equality constraints...: 0
Total number of inequality constraints...: 2
  inequality constraints with only lower bounds: 2
  inequality constraints with lower and upper bounds: 0
  inequality constraints with only upper bounds: 0
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
 0  9.9999900e-03  9.70e-01  1.29e+00 -1.0 0.00e+00 - 0.00e+00 0.00e+00 0
 1  1.1575770e-01  6.07e-01  2.14e+00 -1.0 3.54e-01 - 3.74e-01 2.90e-01f 1
 2  4.1080064e-01  0.00e+00 2.30e-01 -1.0 2.95e-01 - 8.07e-01 1.00e+00f 1
 3  4.2831409e-01  0.00e+00 9.99e-01 -1.7 9.12e-01 - 4.43e-01 5.00e-01f 2
 4  3.8684538e-01  0.00e+00 5.50e-01 -1.7 1.22e+00 - 5.86e-01 2.46e-01f 2
 5  3.6841155e-01  0.00e+00 4.90e-01 -1.7 8.59e-01 - 5.32e-01 1.25e-01f 4
 6  3.5762409e-01  0.00e+00 4.80e-01 -1.7 4.63e+00 - 2.27e-01 3.15e-03f 8
 7  3.5745122e-01  0.00e+00 4.61e-01 -1.7 3.97e+00 - 1.84e-01 2.29e-04f 12
 8  3.5745086e-01  0.00e+00 4.70e-01 -1.7 5.40e+00 - 6.04e-01 6.00e-07f 20
 9  3.5745095e-01  0.00e+00 5.19e-01 -1.7 2.01e+00 - 2.36e-01 3.22e-07f 22
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
10  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.90e+00 - 2.38e-01 2.03e-07f 23
11  3.5745086e-01  0.00e+00 8.71e-01 -1.7 5.71e+00 - 2.51e-01 5.21e-09f 27
12  3.5745086e-01  0.00e+00 5.05e-01 -1.7 1.65e+00 - 5.98e-01 1.93e-09f 28
13  4.4895064e-01  0.00e+00 8.71e-01 -1.7 1.08e+00 - 3.68e-01 1.77e-01w 1
14  4.3594343e-01  0.00e+00 6.56e-01 -1.7 1.97e+00 - 8.44e-01 3.12e-01w 1
15  3.7675014e-01  0.00e+00 6.17e-01 -1.7 7.15e-01 - 2.28e-01 1.56e-01w 1
16  3.5745086e-01  0.00e+00 8.47e-01 -1.7 5.56e+00 - 3.68e-01 1.32e-09f 27
17  3.5745086e-01  0.00e+00 5.10e-01 -1.7 1.82e+00 - 7.90e-01 2.17e-10f 31
18  3.5745086e-01  0.00e+00 4.85e-01 -1.7 1.24e+00 - 3.69e-01 1.07e-10f 32
19  3.5745086e-01  0.00e+00 8.71e-01 -1.7 5.13e+00 - 2.23e-01 2.87e-12f 38
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
20  3.5745086e-01  0.00e+00 5.21e-01 -1.7 1.65e+00 - 6.92e-01 4.68e-13f 40
21  3.5745086e-01  0.00e+00 4.99e-01 -1.7 1.01e+00 - 3.83e-01 1.48e-13f 41
22  3.5745086e-01  0.00e+00 8.71e-01 -1.7 1.71e+00 - 2.24e-01 1.10e-14f 46
23  3.5745086e-01  0.00e+00 5.10e-01 -1.7 1.69e+00 - 6.62e-01 1.43e-14f 45
24  3.5745086e-01  0.00e+00 8.46e-01 -1.7 1.05e+00 - 3.76e-01 9.67e-15f 45
25  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.83e+00 - 7.75e-01 6.61e-15f 46
26  4.9348017e-01  0.00e+00 8.56e-01 -1.7 1.76e+00 - 3.66e-01 2.42e-01w 1
27  4.4073908e-01  0.00e+00 6.84e-01 -1.7 2.02e+00 - 8.58e-01 3.60e-01w 1
28  3.7286193e-01  0.00e+00 6.42e-01 -1.7 8.44e-01 - 2.43e-01 1.55e-01w 1
29  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.08e+00 - 3.68e-01 6.87e-15f 45
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
30  3.5745086e-01  0.00e+00 5.09e-01 -1.7 1.82e+00 - 7.86e-01 3.32e-15f 47
31  3.5745086e-01  0.00e+00 8.46e-01 -1.7 1.25e+00 - 3.78e-01 3.36e-15f 47
32  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.83e+00 - 7.76e-01 3.31e-15f 47
33  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.27e+00 - 3.66e-01 6.97e-15f 46
34  3.5745086e-01  0.00e+00 5.09e-01 -1.7 1.82e+00 - 7.86e-01 6.64e-15f 46
35  3.5745086e-01  0.00e+00 8.46e-01 -1.7 1.25e+00 - 3.69e-01 6.74e-15f 46
36  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.83e+00 - 7.76e-01 3.31e-15f 47
37  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.27e+00 - 3.66e-01 3.48e-15f 47
38  3.5745086e-01  0.00e+00 5.09e-01 -1.7 1.82e+00 - 7.85e-01 3.32e-15f 47
39  4.9804071e-01  0.00e+00 8.55e-01 -1.7 1.25e+00 - 3.69e-01 2.38e-01w 1
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
40  4.3996743e-01  0.00e+00 6.79e-01 -1.7 2.02e+00 - 8.39e-01 3.55e-01w 1
41  3.7150575e-01  0.00e+00 6.35e-01 -1.7 8.56e-01 - 2.35e-01 1.60e-01w 1
42  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.09e+00 - 3.69e-01 3.38e-15f 46
43  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.83e+00 - 7.77e-01 1.65e-15f 48
44  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.66e-01 3.47e-15f 47
45  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.84e-01 3.32e-15f 47
46  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.25e+00 - 3.69e-01 3.38e-15f 47
47  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.78e-01 3.31e-15f 47
48  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 6.92e-15f 46
49  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.84e-01 6.63e-15f 46
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
50  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.25e+00 - 3.69e-01 6.78e-15f 46
51  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.78e-01 3.31e-15f 47
52  4.9469217e-01  0.00e+00 8.55e-01 -1.7 1.26e+00 - 3.67e-01 2.43e-01w 1
53  4.4075258e-01  0.00e+00 6.83e-01 -1.7 2.02e+00 - 8.53e-01 3.61e-01w 1
54  3.7227290e-01  0.00e+00 6.41e-01 -1.7 8.54e-01 - 2.41e-01 1.56e-01w 1
55  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.77e+00 - 3.67e-01 3.46e-15f 46
56  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.83e-01 3.32e-15f 47
57  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.25e+00 - 3.68e-01 3.39e-15f 47
58  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.78e-01 1.66e-15f 48
59  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 3.45e-15f 47
```

```
iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
60  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.82e+00 - 7.81e-01 3.32e-15f 47
61  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.40e-15f 47
62  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.79e-01 3.31e-15f 47
63  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 3.46e-15f 47
64  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.83e-01 1.66e-15f 48
65  4.928490e-01  0.00e+00 8.55e-01 -1.7 1.26e+00 - 3.68e-01 2.39e-01w 1
66  4.4021140e-01  0.00e+00 6.80e-01 -1.7 2.02e+00 - 8.42e-01 3.57e-01w 1
67  3.7175141e-01  0.00e+00 6.37e-01 -1.7 8.55e-01 - 2.37e-01 1.59e-01w 1
68  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.69e+00 - 3.68e-01 3.40e-15f 46
69  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.79e-01 3.32e-15f 47
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iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
70  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 3.45e-15f 47
71  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 3.31e-15f 47
72  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.81e-15f 46
73  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.79e-01 6.62e-15f 46
74  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 6.80e-15f 46
75  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 3.31e-15f 47
76  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.41e-15f 47
77  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.79e-01 3.31e-15f 47
78  4.9384981e-01  0.00e+00 8.55e-01 -1.7 1.26e+00 - 3.67e-01 2.42e-01w 1
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iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
79  4.4858664e-01  0.00e+00 6.82e-01 -1.7 2.02e+00 - 8.58e-01 3.60e-01w 1
80  3.7211277e-01  0.00e+00 6.40e-01 -1.7 8.54e-01 - 2.40e-01 1.57e-01w 1
81  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.74e+00 - 3.67e-01 3.44e-15f 46
82  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 1.66e-15f 48
83  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.41e-15f 47
84  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.78e-01 3.31e-15f 47
85  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 3.44e-15f 47
86  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 3.31e-15f 47
87  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.82e-15f 46
88  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.80e-01 6.62e-15f 46
89  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 6.87e-15f 46
```

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iter   objective    inf_pr  inf_du lg(mu)  ||d|| lg(rg) alpha_du alpha_pr ls
90  3.5745086e-01  0.00e+00 5.06e-01 -1.7 1.82e+00 - 7.82e-01 3.31e-15f 47
91  4.9259988e-01  0.00e+00 8.55e-01 -1.7 1.26e+00 - 3.68e-01 2.40e-01w 1
92  4.4027800e-01  0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.45e-01 3.58e-01w 1
93  3.7186951e-01  0.00e+00 6.36e-01 -1.7 8.55e-01 - 2.38e-01 1.59e-01w 1
94  3.5745086e-01  0.00e+00 8.45e-01 -1.7 5.70e+00 - 3.68e-01 3.41e-15f 46
95  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.80e-01 3.31e-15f 47
96  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.67e-01 3.45e-15f 47
97  3.5745086e-01  0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 1.66e-15f 48
98  3.5745086e-01  0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
99  3.5745086e-01  0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.80e-01 3.31e-15f 47
```

[illegible]

iter	objective	inf_pr	inf_du	inf_lm(mu)	d	lg(rg)	alpha	du	ls
100	4.4043258e-01	0.00e00	6.81e-01	-1.7 2.02e+00	-	8.47e-01	3.59e-01w	1	
101	3.71966720e-01	0.00e00	6.39e-01	-1.7 8.55e-01	-	2.38e-01	1.58e-01w	1	
102	3.5745886e-01	0.00e00	8.45e-01	-1.7 5.72e+00	-	3.68e-01	6.85e-15f	45	
103	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
104	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
105	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
106	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
107	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	1.66e-15f	48	
108	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
109	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
iter	objective	inf_pr	inf_du	inf_lm(mu)	d	lg(rg)	alpha	du	ls
110	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
111	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
112	4.9310224e-01	0.00e00	5.55e-01	-1.7 1.26e+00	-	3.68e-01	2.41e-01w	1	
113	4.4043330e-01	0.00e00	6.81e-01	-1.7 2.02e+00	-	8.47e-01	3.59e-01w	1	
114	3.71966720e-01	0.00e00	6.39e-01	-1.7 8.55e-01	-	2.38e-01	1.58e-01w	1	
115	3.5745886e-01	0.00e00	8.45e-01	-1.7 5.72e+00	-	3.68e-01	6.85e-15f	45	
116	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	6.35e-15f	46	
117	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	6.85e-15f	46	
118	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
119	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
iter	objective	inf_pr	inf_du	inf_lm(mu)	d	lg(rg)	alpha	du	ls
120	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
121	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
122	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
123	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	6.85e-15f	46	
124	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	6.85e-15f	46	
125	4.9310071e-01	0.00e00	5.55e-01	-1.7 1.26e+00	-	3.68e-01	2.41e-01w	1	
126	4.4043272e-01	0.00e00	6.81e-01	-1.7 2.02e+00	-	8.47e-01	3.59e-01w	1	
127	3.7196691e-01	0.00e00	6.39e-01	-1.7 8.55e-01	-	2.38e-01	1.58e-01w	1	
128	3.5745886e-01	0.00e00	8.45e-01	-1.7 5.72e+00	-	3.68e-01	6.85e-15f	45	
129	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
iter	objective	inf_pr	inf_du	inf_lm(mu)	d	lg(rg)	alpha	du	ls
130	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68e-01	3.42e-15f	47	
131	3.5745886e-01	0.00e00	5.07e-01	-1.7 1.82e+00	-	7.81e-01	3.31e-15f	47	
132	3.5745886e-01	0.00e00	8.45e-01	-1.7 1.26e+00	-	3.68			

iter	objective	inf	pr	inf	lg(mu)	d	lg(lr)	alpha	alpha	pr
387	3.5745886e-01	0.00e+00	8.45e-01	-1.7.572e+00	-	3.68e-01	3.42e-15f	47		
388	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
391	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
393	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.26e+00	-	7.81e-01	3.31e-15f	47		
394	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
395	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
396	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	3.42e-15f	47		
397	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
398	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
399	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.26e+00	-	7.81e-01	3.31e-15f	47		
iter	objective	inf <td>pr<td>inf<td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td></td></td>	pr <td>inf<td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td></td>	inf <td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td>	lg(mu) <th> d </th> <th>lg(lr)</th> <th>alpha</th> <th>alpha</th> <th>pr</th>	d	lg(lr)	alpha	alpha	pr
400	4.9310145e-01	0.00e+00	8.55e-01	-1.7.1.26e+00	-	3.68e-01	2.41e-01w	1		
401	4.9432862e-01	0.00e+00	6.81e-01	-1.7.2.02e+00	-	8.47e-01	3.59e-01w	1		
402	3.7196704e-01	0.00e+00	6.39e-01	-1.7.855e-01	-	2.38e-01	1.58e-01w	1		
403	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	1		
404	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.82e+00	-	7.81e-01	6.85e-15f	1		
405	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	1		
406	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
407	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.26e+00	-	3.68e-01	3.42e-15f	47		
408	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
409	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
iter	objective	inf <td>pr<td>inf<td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td></td></td>	pr <td>inf<td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td></td>	inf <td>lg(mu)<th> d </th><th>lg(lr)</th><th>alpha</th><th>alpha</th><th>pr</th></td>	lg(mu) <th> d </th> <th>lg(lr)</th> <th>alpha</th> <th>alpha</th> <th>pr</th>	d	lg(lr)	alpha	alpha	pr
410	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	3.42e-15f	47		
411	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
412	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
413	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	3.31e-15f	47		
414	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.26e+00	-	3.68e-01	6.85e-15f	47		
415	3.5745886e-01	0.00e+00	5.67e-01	-1.7.1.82e+00	-	7.81e-01	6.85e-15f	47		
416	4.9310141e-01	0.00e+00	8.55e-01	-1.7.1.26e+00	-	3.68e-01	2.41e-01w	1		
417	4.9432862e-01	0.00e+00	6.81e-01	-1.7.2.02e+00	-	8.47e-01	3.59e-01w	1		
418	3.7196704e-01	0.00e+00	6.39e-01	-1.7.855e-01	-	2.38e-01	1.58e-01w	1		
419	3.5745886e-01	0.00e+00	8.45e-01	-1.7.1.82e+00	-	7.81e-01	6.85e-15f	47		

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iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
460 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
461 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
462 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
463 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
464 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
465 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
466 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
467 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
468 4.9310137e-01 0.00e+00 6.81e-01 -1.7 1.26e+00 - 3.68e-01 2.41e-01w 1
469 4.4043285e-01 0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.47e-01 3.59e-01w 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
470 3.7196703e-01 0.00e+00 6.39e-01 -1.7 8.55e-01 - 2.38e-01 1.58e-01w 1
471 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.72e+00 - 3.68e-01 6.85e-15f 45
472 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
473 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
474 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
475 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
476 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 1.66e-15f 48
477 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
478 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
479 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
480 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
481 4.9310135e-01 0.00e+00 6.85e-01 -1.7 1.26e+00 - 3.68e-01 2.41e-01w 1
482 4.4043284e-01 0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.47e-01 3.59e-01w 1
483 3.7196703e-01 0.00e+00 6.39e-01 -1.7 8.55e-01 - 2.38e-01 1.58e-01w 1
484 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.72e+00 - 3.68e-01 6.85e-15f 45
485 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
486 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
487 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
488 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
489 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
490 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
491 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
492 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
493 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
494 4.9310137e-01 0.00e+00 6.85e-01 -1.7 1.26e+00 - 3.68e-01 2.41e-01w 1
495 4.4043285e-01 0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.47e-01 3.59e-01w 1
496 3.7196703e-01 0.00e+00 6.39e-01 -1.7 8.55e-01 - 2.38e-01 1.58e-01w 1
497 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.72e+00 - 3.68e-01 6.85e-15f 45
498 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
499 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
```

```
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
500 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
501 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
502 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
503 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
504 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
505 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
506 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
507 4.9310136e-01 0.00e+00 6.85e-01 -1.7 1.26e+00 - 3.68e-01 2.41e-01w 1
508 4.4043284e-01 0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.47e-01 3.59e-01w 1
509 3.7196703e-01 0.00e+00 6.39e-01 -1.7 8.55e-01 - 2.38e-01 1.58e-01w 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
510 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.72e+00 - 3.68e-01 3.42e-15f 46
511 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 1.66e-15f 48
512 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
513 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
514 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
515 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
516 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
517 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 6.33e-15f 46
518 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
519 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
520 4.9310136e-01 0.00e+00 6.85e-01 -1.7 1.26e+00 - 3.68e-01 2.41e-01w 1
521 4.4043285e-01 0.00e+00 6.81e-01 -1.7 2.02e+00 - 8.47e-01 3.59e-01w 1
522 3.7196703e-01 0.00e+00 6.39e-01 -1.7 8.55e-01 - 2.38e-01 1.58e-01w 1
523 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.72e+00 - 3.68e-01 3.42e-15f 46
524 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
525 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
526 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
527 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 6.85e-15f 46
528 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 3.31e-15f 47
529 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 1.71e-15h 48
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
530 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.82e+00 - 7.81e-01 1.66e-15f 48
531 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.26e+00 - 3.68e-01 3.42e-15f 47
532 3.5745886e-01 0.00e+00 7.79e-01 -1.7 1.82e+00 - 7.81e-01 1.66e-15f 48
533 2.2000609e+00 0.00e+00 6.43e-01 -1.7 9.31e+00 - 1.777e-02 3.55e-01w 1
534 1.2657019e+00 0.00e+00 9.46e-01 -1.7 1.70e+00 - 5.68e-01 1.00e+00w 1
535 1.2982576e+00 0.00e+00 1.06e+00 -1.7 2.33e+00 - 3.59e-01 1.00e+00w 1
536 3.5745886e-01 0.00e+00 5.21e-01 -1.7 2.00e+00 - 8.77e-02 3.19e-15f 50
537 3.5745886e-01 0.00e+00 8.54e-01 -1.7 1.75e+00 - 3.83e-01 3.02e-15f 49
538 3.5745886e-01 0.00e+00 4.95e-01 -1.7 1.90e+00 - 7.31e-01 3.19e-15f 47
539 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.44e+00 - 3.29e-01 4.50e-15f 47
```

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iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
540 3.5745886e-01 0.00e+00 5.19e-01 -1.7 1.80e+00 - 8.16e-01 1.68e-15f 48
541 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.20e+00 - 3.78e-01 1.50e-15f 48
542 3.5745886e-01 0.00e+00 4.97e-01 -1.7 1.85e+00 - 7.39e-01 1.64e-15f 48
543 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.33e+00 - 3.55e-01 1.99e-15h 48
544 3.5745886e-01 0.00e+00 5.18e-01 -1.7 1.81e+00 - 8.18e-01 3.35e-15f 47
545 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.20e+00 - 3.79e-01 3.04e-15h 47
546 4.7168079e-01 0.00e+00 6.14e-01 -1.7 1.84e+00 - 7.47e-01 2.31e-01w 1
547 3.8388856e-01 0.00e+00 5.96e-01 -1.7 8.35e-01 - 5.13e-01 1.43e-01w 1
548 1.8640245e+00 0.00e+00 9.21e-01 -1.7 6.82e+00 - 2.65e-01 3.10e-01w 1
549 3.5745886e-01 0.00e+00 4.99e-01 -1.7 1.62e+00 - 7.47e-01 1.64e-15f 47
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
550 3.5745886e-01 0.00e+00 4.77e-01 -1.7 1.32e+00 - 3.57e-01 1.94e-15h 48
551 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.86e+00 - 1.22e-02 3.59e-01w 1
552 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.62e+00 - 6.03e-01 3.73e-15f 47
553 3.5745886e-01 0.00e+00 8.47e-01 -1.7 1.24e+00 - 3.26e-01 4.17e-15f 47
554 3.5745886e-01 0.00e+00 4.94e-01 -1.7 1.86e+00 - 7.42e-01 1.62e-15f 48
555 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.32e+00 - 3.55e-01 2.00e-15h 48
556 3.5745886e-01 0.00e+00 5.21e-01 -1.7 1.80e+00 - 8.15e-01 1.68e-15f 48
557 3.5745886e-01 0.00e+00 5.00e-01 -1.7 1.23e+00 - 3.70e-01 1.67e-15h 48
558 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.18e+00 - 2.11e-02 6.89e-16f 50
559 4.8682433e-01 0.00e+00 6.29e-01 -1.7 1.69e+00 - 6.52e-01 2.52e-01w 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
560 3.9057768e-01 0.00e+00 6.24e-01 -1.7 1.80e+00 - 4.82e-01 1.34e-01w 1
561 1.0733957e+00 0.00e+00 7.86e-01 -1.7 7.96e+00 - 7.72e-02 2.74e-01w 1
562 3.5745886e-01 0.00e+00 5.08e-01 -1.7 1.72e+00 - 6.52e-01 3.58e-15f 46
563 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.05e+00 - 3.74e-01 4.88e-15f 46
564 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.80e-01 1.05e-15f 48
565 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.25e+00 - 3.67e-01 1.69e-15h 48
566 3.5745886e-01 0.00e+00 5.08e-01 -1.7 1.82e+00 - 7.82e-01 1.66e-15f 48
567 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.25e+00 - 3.69e-01 1.70e-15h 48
568 3.5745886e-01 0.00e+00 5.07e-01 -1.7 1.83e+00 - 7.80e-01 1.66e-15f 48
569 3.5745886e-01 0.00e+00 4.86e-01 -1.7 1.26e+00 - 3.67e-01 8.59e-16h 49
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
570 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.12e+00 - 2.24e-02 1.76e-16h 52
571 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.64e+00 - 6.60e-01 4.68e-16h 50
572 4.6187602e-01 0.00e+00 8.73e-01 -1.7 1.10e+00 - 3.58e-01 2.05e-01w 1
573 4.3262258e-01 0.00e+00 6.49e-01 -1.7 2.03e+00 - 7.65e-01 3.19e-01w 1
574 3.6221235e-01 0.00e+00 5.77e-01 -1.7 8.31e-01 - 1.90e-01 2.21e-01w 1
575 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.84e+00 - 3.58e-01 9.00e-17h 51
576 3.5745886e-01 0.00e+00 4.95e-01 -1.7 1.85e+00 - 7.42e-01 1.63e-15f 48
577 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.31e+00 - 3.57e-01 3.92e-15f 47
578 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.80e+00 - 8.29e-01 3.35e-15f 47
579 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.18e+00 - 3.62e-01 2.95e-15h 47
```



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iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
588 3.5745886e-01 0.00e+00 4.97e-01 -1.7 1.85e+00 - 7.40e-01 1.64e-15f 48
589 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.85e+00 - 3.55e-01 1.99e-15h 48
590 3.5745886e-01 0.00e+00 5.10e-01 -1.7 1.81e+00 - 8.20e-01 8.37e-16f 49
591 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.20e+00 - 3.79e-01 1.51e-15h 48
592 3.5745886e-01 0.00e+00 4.90e-01 -1.7 1.84e+00 - 7.45e-01 1.64e-15f 48
593 5.1639400e-01 0.00e+00 5.56e-01 -1.7 1.32e+00 - 3.55e-01 1.74e-01w 1
594 4.4516436e-01 0.00e+00 6.99e-01 -1.7 2.02e+00 - 7.66e-01 3.80e-01w 1
595 3.4643244e-01 0.00e+00 6.22e-01 -1.7 1.03e+00 - 1.85e-01 2.55e-01w 1
596 3.5745886e-01 0.00e+00 8.45e-01 -1.7 6.62e+00 - 3.55e-01 1.95e-15h 48
597 3.5745886e-01 0.00e+00 5.17e-01 -1.7 1.81e+00 - 8.15e-01 1.67e-15f 48
598 4.8022558e-01 0.00e+00 8.54e-01 -1.7 1.22e+00 - 3.75e-01 2.22e-01w 1
599 4.3776233e-01 0.00e+00 6.67e-01 -1.7 2.03e+00 - 8.02e-01 3.42e-01w 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
600 3.6879724e-01 0.00e+00 6.15e-01 -1.7 8.63e-01 - 2.10e-01 1.75e-01w 1
601 3.5745886e-01 0.00e+00 8.46e-01 -1.7 5.36e+00 - 3.75e-01 1.50e-15h 47
602 3.5745886e-01 0.00e+00 5.02e-01 -1.7 1.84e+00 - 7.59e-01 1.05e-15f 48
603 3.5745886e-01 0.00e+00 4.90e-01 -1.7 1.30e+00 - 3.61e-01 1.85e-15h 48
604 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.00e+00 - 2.26e-01 3.57e-16f 51
605 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.63e+00 - 6.19e-01 1.85e-15f 48
606 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.21e+00 - 3.31e-01 1.80e-15h 48
607 3.5745886e-01 0.00e+00 4.94e-01 -1.7 1.86e+00 - 7.42e-01 8.13e-16f 49
608 3.5745886e-01 0.00e+00 6.45e-01 -1.7 1.32e+00 - 3.56e-01 9.94e-16h 49
609 3.5745886e-01 0.00e+00 5.21e-01 -1.7 1.80e+00 - 6.20e-01 6.39e-16f 49
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
610 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.22e+00 - 3.73e-01 1.61e-15h 48
611 4.7166822e-01 0.00e+00 6.12e-01 -1.7 1.05e+00 - 7.30e-01 2.30e-01w 1
612 3.8355677e-01 0.00e+00 5.94e-01 -1.7 8.25e-01 - 5.80e-01 1.45e-01w 1
613 1.0670637e+00 0.00e+00 9.22e-01 -1.7 6.57e+00 - 2.90e-01 3.23e-01w 1
614 3.5745886e-01 0.00e+00 4.96e-01 -1.7 1.68e+00 - 7.30e-01 1.64e-15f 47
615 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.22e+00 - 3.55e-01 2.00e-15h 48
616 3.5745886e-01 0.00e+00 5.19e-01 -1.7 1.81e+00 - 8.25e-01 1.67e-15f 48
617 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.19e+00 - 3.81e-01 1.99e-15f 47
618 3.5745886e-01 0.00e+00 4.97e-01 -1.7 1.85e+00 - 7.42e-01 1.27e-15f 47
619 3.5745886e-01 0.00e+00 4.76e-01 -1.7 1.32e+00 - 3.56e-01 1.97e-15h 48
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iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
620 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.05e+00 - 2.27e-01 7.20e-16f 50
621 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.62e+00 - 5.97e-01 1.87e-15f 48
622 3.5745886e-01 0.00e+00 8.47e-01 -1.7 1.26e+00 - 3.23e-01 2.15e-15h 48
623 3.5745886e-01 0.00e+00 7.94e-01 -1.7 1.06e+00 - 7.42e-01 8.11e-16f 49
624 2.0971183e+00 0.00e+00 6.22e-01 -1.7 7.00e+00 - 9.95e-02 4.32e-01w 1
625 1.1249240e+00 0.00e+00 9.37e-01 -1.7 1.67e+00 - 5.15e-01 1.00e+00w 1
626 1.1526100e+00 0.00e+00 1.06e+00 -1.7 1.89e+00 - 3.89e-01 1.00e+00w 1
627 3.5745886e-01 0.00e+00 5.21e-01 -1.7 1.61e+00 - 9.95e-02 1.92e-16f 51
628 3.5745886e-01 0.00e+00 8.48e-01 -1.7 1.69e+00 - 3.24e-01 6.49e-16h 51
629 3.5745886e-01 0.00e+00 4.94e-01 -1.7 1.87e+00 - 7.20e-01 1.62e-15f 48
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
630 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.38e+00 - 3.45e-01 4.31e-15f 47
631 3.5745886e-01 0.00e+00 7.75e-01 -1.7 1.80e+00 - 8.20e-01 1.68e-15f 48
632 3.5745886e-01 0.00e+00 5.09e-01 -1.7 8.02e+00 - 2.30e-01 1.95e-16f 52
633 3.5745886e-01 0.00e+00 8.69e-01 -1.7 2.66e+00 - 2.83e-01 1.57e-15f 50
634 3.5745886e-01 0.00e+00 5.00e-01 -1.7 1.62e+00 - 5.75e-01 3.73e-15f 47
635 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.03e+00 - 3.80e-01 4.94e-15f 46
636 3.5745886e-01 0.00e+00 5.16e-01 -1.7 1.82e+00 - 8.18e-01 1.67e-15f 48
637 4.7231130e-01 0.00e+00 8.155e-01 -1.7 1.10e+00 - 3.70e-01 2.10e-01w 1
638 4.3652048e-01 0.00e+00 6.62e-01 -1.7 2.03e+00 - 7.89e-01 3.34e-01w 1
639 3.6788490e-01 0.00e+00 6.09e-01 -1.7 8.54e-01 - 2.11e-01 1.82e-01w 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
640 3.5745886e-01 0.00e+00 8.46e-01 -1.7 5.23e+00 - 3.78e-01 1.51e-15h 47
641 3.5745886e-01 0.00e+00 5.00e-01 -1.7 1.02e+00 - 7.52e-01 3.25e-15f 47
642 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.30e+00 - 3.60e-01 3.78e-15f 47
643 3.5745886e-01 0.00e+00 5.15e-01 -1.7 1.81e+00 - 8.09e-01 1.67e-15f 48
644 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.22e+00 - 3.76e-01 3.13e-15f 47
645 3.5745886e-01 0.00e+00 5.01e-01 -1.7 1.84e+00 - 7.56e-01 1.64e-15f 48
646 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.30e+00 - 3.60e-01 1.87e-15f 48
647 3.5745886e-01 0.00e+00 5.14e-01 -1.7 1.81e+00 - 8.05e-01 3.33e-15f 47
648 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.22e+00 - 3.75e-01 6.34e-15f 46
649 3.5745886e-01 0.00e+00 5.02e-01 -1.7 1.84e+00 - 7.59e-01 6.50e-15f 46
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
650 5.0645545e-01 0.00e+00 8.35e-01 -1.7 2.70e+00 - 3.61e-01 2.60e-01w 1
651 4.4315825e-01 0.00e+00 6.96e-01 -1.7 2.02e+00 - 8.79e-01 3.76e-01w 1
652 3.7188784e-01 0.00e+00 6.54e-01 -1.7 8.63e-01 - 2.46e-01 1.57e-01w 1
653 3.5745886e-01 0.00e+00 8.45e-01 -1.7 5.89e+00 - 3.61e-01 7.39e-15f 45
654 3.5745886e-01 0.00e+00 5.15e-01 -1.7 1.81e+00 - 8.01e-01 3.33e-15f 47
655 3.5745886e-01 0.00e+00 4.21e-01 -1.7 1.22e+00 - 3.74e-01 1.60e-15h 48
656 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.17e+00 - 2.22e-01 6.95e-16f 50
657 3.5745886e-01 0.00e+00 5.18e-01 -1.7 1.66e+00 - 6.88e-01 1.82e-15f 48
658 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.00e+00 - 3.87e-01 1.12e-15h 48
659 3.5745886e-01 0.00e+00 4.99e-01 -1.7 1.84e+00 - 7.48e-01 1.64e-15f 48
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iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
660 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.30e+00 - 3.59e-01 1.80e-15f 48
661 3.5745886e-01 0.00e+00 5.16e-01 -1.7 1.81e+00 - 8.13e-01 1.67e-15f 48
662 3.5745886e-01 0.00e+00 8.46e-01 -1.7 1.21e+00 - 3.70e-01 3.00e-15f 47
663 4.7168785e-01 0.00e+00 6.15e-01 -1.7 1.04e+00 - 7.52e-01 2.31e-01w 1
664 3.4940895e-01 0.00e+00 5.07e-01 -1.7 6.39e-01 - 5.12e-01 1.45e-01w 1
665 1.0675937e+00 0.00e+00 9.21e-01 -1.7 6.52e+00 - 3.04e-01 3.26e-01w 1
666 3.5745886e-01 0.00e+00 5.00e-01 -1.7 1.67e+00 - 7.52e-01 3.28e-15f 46
667 3.5745886e-01 0.00e+00 8.72e-01 -1.7 1.31e+00 - 3.59e-01 1.90e-15f 48
668 3.5745886e-01 0.00e+00 8.71e-01 -1.7 5.07e+00 - 2.27e-01 7.16e-16f 50
669 3.5745886e-01 0.00e+00 5.20e-01 -1.7 1.63e+00 - 6.10e-01 3.72e-15f 47
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
670 3.5745886e-01 0.00e+00 8.47e-01 -1.7 1.12e+00 - 3.29e-01 8.00e-15f 46
671 3.5745886e-01 0.00e+00 4.94e-01 -1.7 1.86e+00 - 7.42e-01 6.50e-15f 46
672 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.32e+00 - 3.55e-01 7.90e-15f 46
673 3.5745886e-01 0.00e+00 5.21e-01 -1.7 1.80e+00 - 8.17e-01 6.71e-15f 46
674 3.5745886e-01 0.00e+00 8.45e-01 -1.7 1.23e+00 - 3.71e-01 6.57e-15f 46
675 3.5745886e-01 0.00e+00 4.96e-01 -1.7 1.85e+00 - 7.37e-01 3.20e-15f 47
676 5.2002749e-01 0.00e+00 8.56e-01 -1.7 1.32e+00 - 3.57e-01 2.81e-01w 1
677 4.4579884e-01 0.00e+00 6.99e-01 -1.7 2.02e+00 - 7.12e-01 3.94e-01w 1
678 3.4379745e-01 0.00e+00 5.68e-01 -1.7 1.12e+00 - 1.60e-01 3.14e-01w 1
679 5.2002749e-01 0.00e+00 8.42e-01 -1.7 1.71e+00 - 2.81e-01 2.81e-01w 30
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
680 4.6351510e-01 0.00e+00 7.19e-01 -1.7 1.85e+00 - 8.12e-01 4.29e-01f 1
681 3.8464200e-01 0.00e+00 6.90e-01 -1.7 8.54e-01 - 4.05e-01 1.29e-01f 1
682 4.3396775e-01 0.00e+00 8.92e-01 -1.7 9.66e+00 - 1.36e-01 5.29e-02f 3
683 5.0049567e-01 0.00e+00 7.56e-01 -1.7 1.36e+01 - 4.04e-02 2.90e-02f 5
684 4.8918020e-01 0.00e+00 6.91e-01 -1.7 2.49e+00 - 3.25e-01 7.41e-02f 4
685 4.8436994e-01 0.00e+00 7.62e-01 -1.7 1.89e+00 - 6.52e-01 5.94e-02f 4
686 4.8048156e-01 0.00e+00 9.12e-01 -1.7 1.49e+00 - 5.34e-01 5.60e-02f 4
687 4.6104046e-01 0.00e+00 7.92e-01 -1.7 1.70e+00 - 5.42e-01 3.05e-02f 5
688 4.5821240e-01 0.00e+00 8.01e-01 -1.7 1.96e+00 - 4.47e-01 4.06e-01f 8
689 4.5802144e-01 0.00e+00 7.80e-01 -1.7 1.89e+00 - 7.02e-01 4.26e-04f 11
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha du alpha_pr ls
690 4.5700232e-01 0.00e+00 8.12e-01 -1.7 1.43e+00 - 5.63e-01 2.00e-04f 12
691 4.5794040e-01 0.00e+00 7.93e-01 -1.7 1.81e+00 - 5.89e-01 1.11e-04f 13
692 8.8770636e-01 0.00e+00 9.41e-01 -1.7 1.83e+00 - 4.75e-01 4.85e-01w 1
693 6.1637297e-01 0.00e+00 9.75e-01 -1.7 2.20e+00 - 4.82e-01 7.73e-01w 1
694 1.2155239e+00 0.00e+00 8.30e-01 -1.7 3.40e+00 - 2.17e-01 7.75e-01w 1
695 3.6583080e-01 0.00e+00 7.97e-01 -1.7 2.19e+00 - 2.24e-01 1.00e+00f 1
696 3.6540704e-01 0.00e+00 8.59e-01 -1.7 5.12e+00 - 1.54e-01 4.40e-04f 11
697 3.6583337e-01 0.00e+00 7.75e-01 -1.7 1.79e+00 - 4.94e-01 5.00e-04f 10
698 3.6580549e-01 0.00e+00 8.50e-01 -1.7 2.10e+00 - 2.81e-01 4.26e-04f 11
699 3.6574525e-01 0.00e+00 7.97e-01 -1.7 1.00e+00 - 7.72e-01 2.40e-04f 11
```

```

iter   objective    inf_pr   inf_du lg(mu)  ||d||  lg(rg) alpha_du alpha_pr ls
700   3.6571135e-01  0.00e+00  8.52e-01 -1.7 1.23e+00 - 4.15e-01 1.72e-04f 13
701   3.6567842e-01  0.00e+00  7.77e-01 -1.7 1.02e+00 - 7.22e-01 5.32e-05f 13
702   3.6567155e-01  0.00e+00  7.52e-01 -1.7 1.32e+00 - 3.92e-01 1.26e-05f 15
703   3.6567155e-01  0.00e+00  8.67e-01 -1.7 5.60e+00 - 2.60e-01 9.31e-08f 23
704   3.6567151e-01  0.00e+00  7.97e-01 -1.7 1.55e+00 - 5.15e-01 7.05e-08f 23
705   3.6567151e-01  0.00e+00  8.46e-01 -1.7 1.40e+00 - 3.23e-01 1.66e-08f 25
706   4.9085955e-01  0.00e+00  8.28e-01 -1.7 1.89e+00 - 7.28e-01 2.43e-01w 1
707   3.7978421e-01  0.00e+00  7.85e-01 -1.7 9.12e-01 - 4.81e-01 1.46e-01w 1
708   1.1779463e+00  0.00e+00  9.99e-01 -1.7 3.85e+00 - 8.02e-02 6.32e-01w 1
709   3.6567150e-01  0.00e+00  7.72e-01 -1.7 2.80e+00 - 7.28e-01 7.24e-09f 25
iter   objective    inf_pr   inf_du lg(mu)  ||d||  lg(rg) alpha_du alpha_pr ls
710   3.6567150e-01  0.00e+00  8.52e-01 -1.7 1.25e+00 - 4.15e-01 3.05e-09f 27
711   3.6567150e-01  0.00e+00  7.97e-01 -1.7 1.85e+00 - 7.16e-01 2.38e-10f 31
712   3.6567150e-01  0.00e+00  8.46e-01 -1.7 1.53e+00 - 3.35e-01 7.26e-11f 33
713   3.6567150e-01  0.00e+00  7.73e-01 -1.7 1.88e+00 - 7.22e-01 1.42e-11f 35
714   3.6567150e-01  0.00e+00  7.45e-01 -1.7 1.27e+00 - 4.13e-01 6.02e-12f 36
715   3.6567150e-01  0.00e+00  8.47e-01 -1.7 5.62e+00 - 2.90e-01 3.53e-13f 41
716   3.6567150e-01  0.00e+00  7.97e-01 -1.7 1.49e+00 - 4.86e-01 2.79e-13f 41
717   3.6567150e-01  0.00e+00  8.47e-01 -1.7 1.43e+00 - 3.15e-01 1.39e-13f 42
718   3.6567150e-01  0.00e+00  7.72e-01 -1.7 1.80e+00 - 7.32e-01 1.10e-13f 42
719   4.8687597e-01  0.00e+00  8.64e-01 -1.7 1.25e+00 - 4.15e-01 2.05e-01w 1
iter   objective    inf_pr   inf_du lg(mu)  ||d||  lg(rg) alpha_du alpha_pr ls
720   4.4758087e-01  0.00e+00  8.34e-01 -1.7 1.52e+00 - 6.00e-01 3.57e-01w 1
721   3.7216821e-01  0.00e+00  8.83e-01 -1.7 1.18e+00 - 1.40e-01 3.59e-01w 1
Cannot call restoration phase at point that is almost feasible (violation 0.0000000e+00).
Short in line search due to no other fall back.

Number of Iterations.....: 721

(scaled)                (unscaled)
Objective.....: 3.6507130134048390e-01 3.6507130134048390e-01
Dual infeasibility.....: 7.7167993543273294e-01 7.7167993543273294e-01
Constraint violation....: 0.0000000000000000e+00 0.0000000000000000e+00
Variable bound violation: 0.0000000000000000e+00 0.0000000000000000e+00
Complementarity.....: 2.894006507915121e-02 2.894006507915121e-02
Overall NLP error.....: 7.7167993543273294e-01 7.7167993543273294e-01

Number of objective function evaluations = 24910
Number of objective gradient evaluations = 723
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 24910
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 1
Number of Lagrangian Hessian evaluations = 722
Total seconds in IPOPT = 14.870

```

$$(e) f_0(x_1, x_2) = x_1^2 + 9x_2^2$$

```

julia> Objective(model, Min, x1^2+9x2^2)
x1^2 + 9 x2^2

julia> optimize!(model)
This is Ipopt version 3.14.16, running with linear solver MUMPS 5.7.3.

Number of nonzeros in equality constraint Jacobian...: 0
Number of nonzeros in inequality constraint Jacobian.: 4
Number of nonzeros in Lagrangian Hessian.....: 2

Total number of variables.....: 2
    variables with only lower bounds: 2
    variables with lower and upper bounds: 0
    variables with only upper bounds: 0
Total number of equality constraints.....: 0
Total number of inequality constraints.....: 2
    inequality constraints with only lower bounds: 2
    inequality constraints with lower and upper bounds: 0
    inequality constraints with only upper bounds: 0

iter   objective    inf_pr   inf_du lg(mu)  ||d||  lg(rg) alpha_du alpha_pr ls
0   9.9999980e-04  0.70e-01 1.18e+00 -1.0 0.00e+00 - 0.00e+00 0.00e+00 0
1   1.0506304e-01  6.13e-01 9.48e+00 -1.0 3.61e-01 - 3.65e-02 3.75e-01f 1
2   6.1401585e-01  0.00e+00 6.11e-01 -1.0 2.80e-01 - 6.98e-01 1.00e+00f 1
3   5.534807e-01  0.00e+00 1.22e-01 -1.7 2.37e-01 - 7.92e-01 1.00e+00f 1
4   5.032297e-01  0.00e+00 2.83e-00 -2.5 9.22e-02 - 1.00e+00 1.00e+00f 1
5   5.0009261e-01  0.00e+00 1.50e-09 -3.8 3.89e-02 - 1.00e+00 1.00e+00f 1
6   5.0015298e-01  0.00e+00 1.50e-09 -3.8 7.90e-03 - 1.00e+00 1.00e+00f 1
7   5.0000157e-01  0.00e+00 1.84e-11 -5.7 1.15e-03 - 1.00e+00 1.00e+00f 1
8   4.9999999e-01  0.00e+00 2.51e-14 -8.6 1.66e-05 - 1.00e+00 1.00e+00f 1

Number of Iterations.....: 8

(scaled)                (unscaled)
Objective.....: 4.9999999246437998e-01 4.9999999246437998e-01
Dual infeasibility.....: 2.5091040356528538e-14 2.5091040356528538e-14
Constraint violation....: 0.0000000000000000e+00 0.0000000000000000e+00
Variable bound violation: 0.0000000000000000e+00 0.0000000000000000e+00
Complementarity.....: 2.8500843913959993e-09 2.8500843913959993e-09
Overall NLP error.....: 2.8500843913959993e-09 2.8500843913959993e-09

Number of objective function evaluations = 9
Number of objective gradient evaluations = 9
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 9
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 1
Number of Lagrangian Hessian evaluations = 1
Total seconds in IPOPT = 0.280

```